AI Structural-Break Detector for Statistical Arbitrage

Backtesting & Evaluation Framework

Complete experimental ladder, trade simulation, predictive/agent metrics, robustness, uncertainty, and reporting specification

This document defines how every layer of the system is evaluated under the same chronological information set and execution assumptions. The objective is not to manufacture the best backtest; it is to measure the incremental value of econometric validity, supervised ML, structured point-in-time event features, and the structural-break AI layer.

Headline experiment

Baseline 0 -> Baseline 1 -> Model A -> Model B -> Full System. Every step changes one clearly identified capability so the contribution of each layer can be measured.

Claim boundary

Historical results demonstrate research value under stated assumptions. They are not evidence of deployable alpha or achievable live execution.

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1. Evaluation objectives and hypotheses

Comparison	Question
Baseline 0 vs Baseline 1	Does rolling econometric relationship validity improve threshold-based pair selection?
Baseline 1 vs Model A	Does the supervised XGBoost filter select divergence events with higher realised mean-reversion quality?
Model A vs Model B	Do structured point-in-time event/NLP features improve ranking/calibration and downstream event selection?
Model B vs Full System	Does source-grounded language AI reduce failed opportunities without vetoing too many successful ones?

1.1 Primary evidence types

- Predictive discrimination and probability calibration.
- Event-selection efficiency: success/failure composition across accepted/rejected events.
- Economic outcomes under one fixed simulator and cost assumptions.
- Agent-specific rejection, false-veto, evidence-validity, and abstention metrics.
- Robustness by time, pair, sector, market stress, event type, and implementation costs.

No single headline metric

A higher Sharpe alone is not sufficient. The report must show whether the predictive model is calibrated, whether risk improvement comes from rejecting failures rather than accidental concentration, and whether results survive reasonable cost and period slices.

2. Experimental freeze and chronological partitions

Partition	Working dates	Allowed use
Training	2018-2021 usable events	Fit logistic/XGBoost, training-fold model selection.
Validation	2022-2023 usable events	Feature/event-feature decisions, probability calibration, policy thresholds, agent prompt/policy development on labelled subset.
Final test	2024-2025 usable events	Opened once after method freeze; headline out-of-sample results.
Embargo	At least 10 trading days at boundaries	Prevents labels from earlier partition using future partition prices.

2.1 Freeze manifest

Before the final test is opened, write a machine-readable experiment manifest containing pair universe, label policy, feature schema, Model A/B artefacts, calibrator, probability thresholds, agent prompt/schema versions, decision policy, execution convention, cost assumptions, and code/data commit IDs.

- Changing any headline method after inspecting final-test results creates a new experiment and invalidates the previous “untouched test” claim.
- Point-in-time pair and trailing market features continue to refresh according to the pre-frozen rule during the final test; the predictive model and policy do not retrain.

3. Evaluation ladder and exact acceptance rules

Version	What changes	Automated acceptance rule
Baseline 0 - threshold only	Trade every fresh divergence crossing in the curated pair universe.	Fresh |z| crossing; ignores cointegration/lifecycle validity.
Baseline 1 - valid pairs only	Add econometric eligibility.	Fresh crossing AND pair ACTIVE at event time.
Model A - ML filter	Add quantitative XGBoost selection.	Baseline 1 event AND calibrated P(revert) >= frozen ML threshold.
Model B - ML + event features	Use XGBoost model/features including structured point-in-time event/NLP features.	Baseline 1 event AND Model B calibrated probability >= frozen threshold.
Full system	Add point-in-time event intelligence, agents and deterministic controller.	Only final status CANDIDATE enters simulated trade; REVIEW/NO TRADE/ABSTAIN are withheld.

One change per rung
The headline Model B adds structured point-in-time event/NLP features to XGBoost. The Full System then uses the underlying admissible evidence and structured extraction in the agent/controller layer, so the incremental value of reasoning can be measured separately from tabular event features.

4. Raw signal universe versus ML event universe

Baseline 0 requires threshold crossings even when a pair is not currently cointegrated/ACTIVE. Therefore the backtester maintains a raw signal table distinct from the ML divergence-event table.

Object	Eligibility	Use
raw_divergence_signals	Any curated pair with an estimable current spread/z-score and fresh threshold crossing.	Baseline 0 candidate universe; stores relationship-validity fields even when invalid.
valid_divergence_signals	Raw signal where pair lifecycle=ACTIVE and required econometric rules pass.	Baseline 1 universe.
divergence_events	Valid signal with complete ML features/label-policy metadata.	Model A/B training and scoring.
full_system_events	Scored event with admissible evidence/agent/controller outputs.	Full-system evaluation subset; coverage/abstention explicitly reported.

4.1 Fair-comparison principle

When comparing two adjacent rungs, report results on both the natural eligible universe and a common matched subset where both versions have the required data. This prevents a layer from appearing better merely because missing data removed difficult observations.

5. Simulated pair construction and position sizing

The project simulates a beta-weighted long/short pair using the hedge ratio frozen in the relationship snapshot at signal time. Only pairs with positive, economically sensible beta are eligible for the headline pair trade.

$$w_{A,i} = -\frac{s_i}{1+|\hat{\beta}_i|}, \qquad w_{B,i} = \frac{s_i\hat{\beta}_i}{1+|\hat{\beta}_i|}, \qquad s_i = \text{sign}(z_i)$$

If z>0, leg A is rich relative to B: short A and long beta-weighted B. If z<0, the signs reverse. Absolute weights sum to approximately one for positive beta, so one event uses one unit of gross pair notional in the event-level return calculation.

$$r_{i,t}^{pair} = w_{A,i}r_{A,t} + w_{B,i}r_{B,t} - c_{i,t}$$

Frozen hedge ratio

The entry hedge ratio is frozen for the life of the simulated event. Re-estimating beta after entry would change the position using information not available at the original decision and make comparisons harder to interpret.

6. Signal timing, entry, monitoring, and exit

Step	Convention
Signal	Computed after market close on event day t0 using information through t0.
Entry	Next trading day open (t0+1) using adjusted/open-consistent price series; captures overnight gap after the signal.
Monitoring	At each subsequent close, compute z using the relationship parameters frozen for the signal period.
Convergence trigger	Close reaches |z| <= 0.5.
Adverse trigger	Close reaches same-side |z| >= 3.0.
Time stop	10 trading days after event signal if neither trigger occurs.
Execution of exit	Next available open after a close-based convergence/stop trigger; horizon exit uses the specified next-open convention.
Missing next open	Use explicit missing-execution flag; do not silently substitute a future convenient price.

6.1 Label versus realised trade

The supervised label is defined from the close-based z path. The economic backtest uses a realistic next-open execution convention. Therefore a y=1 event can still have a weak or negative simulated return because of entry/exit gaps and costs. This is desirable: predictive success and economic profitability are related but not identical.

7. Transaction costs, borrow costs, and implementation assumptions

$$C_i = c_{entry} \sum_j |w_{j,i}| + c_{exit} \sum_j |w_{j,i}| + b_i \, |w_i^{short}| \, \frac{H_i}{252}$$

Cost component	Working baseline	Sensitivity
Entry trading cost	5 bps per unit gross notional	2.5 / 5 / 10 / 20 bps.
Exit trading cost	5 bps per unit gross notional	Same grid.
Short borrow	3% annualised on short notional for holding days	0% / 3% / 10% annualised.
Market impact	Not separately modelled with daily public data	Disclose as deployment limitation; high-cost sensitivity partially proxies it.
Hard borrow availability	Not reliably available in public history	Prototype flag/limitation; no claim of live executability.
Dividends/corporate actions	Adjusted price framework where possible	Large corporate events manually checked in case studies.

Cost assumptions are not facts

The baseline values are transparent scenario parameters, not claims about institutional execution. Headline results must include a cost sensitivity table.

8. Event-level return and path metrics

Metric	Definition / purpose
label_success	Whether the predefined mean-reversion target succeeded.
gross_pair_return	Compounded/summed realised pair return between simulated entry and exit before costs.
net_pair_return	Gross pair return minus trading and borrow costs.
holding_days	Number of trading days position is open.
maximum adverse excursion (MAE)	Worst mark-to-market return or same-direction z excursion while open.
maximum favourable excursion (MFE)	Best mark-to-market return or movement toward/beyond equilibrium while open.
exit_reason	CONVERGENCE / ADVERSE_STOP / TIMEOUT / DATA_FAILURE.
overnight_entry_gap	Difference between signal close and next-open execution state; useful error-analysis variable.

8.1 Selection metrics

- Accepted-event success rate and failure rate.
- Mean/median net return per accepted event.
- Distribution and lower-tail quantiles of net returns.
- Rejected-event counterfactual success/failure and return distribution, using the same simulator.
- Candidate retention rate: accepted / eligible events.

9. Portfolio-level simulation

Event-level analysis is primary because it measures selection quality directly. A secondary portfolio simulation converts accepted events into one daily strategy return series for Sharpe/drawdown reporting.

Rule	Working specification
Capital	Normalised initial NAV = 1.0.
Allocation	Equal gross allocation across all open accepted pair trades on each day.
Gross exposure	Total gross exposure capped at 1.0 in the prototype; no leverage.
Concurrent trades	All accepted trades may coexist subject to one open trade per pair; report number of simultaneous positions.
Reweighting	Daily equalisation across open pairs is avoided to reduce artificial turnover; allocate equal notional at entry from available capital and release on exit. A simpler equal-open-position alternative may be used if frozen before test.
Cash	Unused capital earns zero in the prototype; disclose.
Overlapping pairs	Allowed, but report concentration by ticker and sector; no hidden manual exclusions.

9.1 Daily portfolio return

$$r_t^{port} = \sum_{i \in \mathcal{O}_t} a_{i,t}\, r_{i,t}^{pair}$$

Weights a_i,t are determined by the frozen portfolio rule, not optimised using final-test performance.

10. Predictive ML evaluation

Metric	Why reported
ROC-AUC	Threshold-independent ranking quality.
PR-AUC	More informative when success/failure classes are imbalanced.
Precision	Among events accepted by the frozen ML threshold, fraction that are label successes.
Recall	Fraction of all label successes retained by the filter.
F1	Single threshold-based precision/recall balance.
Confusion matrix	Transparent TP/FP/TN/FN counts at the frozen threshold.
Brier score	Probability accuracy/calibration-sensitive score.
Calibration curve	Predicted probability bins versus observed success frequency.

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2$$

Accuracy is not a headline metric

A naive classifier can look good under imbalance. Ranking, calibration, and accepted-event precision/recall are more informative for this application.

11. Calibration and selection-efficiency evaluation

Analysis	Output
Reliability curve	For probability bins, compare mean predicted P(revert) with observed success rate.
Probability buckets	LOW / MEDIUM / HIGH bucket count, success rate, mean net return, failure rate.
Raw vs calibrated Brier	Shows whether Platt calibration improves probability quality.
Top-decile / high-score lift	Success rate in high-score subset relative to Baseline 1 event base rate.
Threshold curve	Validation only: candidate count, precision, recall, success rate and net return across policy thresholds. Test uses frozen threshold.

11.1 Selection efficiency

$$\text{Lift}(B) = \frac{P(y = 1 \mid \hat{p} \in B)}{P(y = 1)}$$

A useful ML layer should rank events so that higher predicted-probability groups have higher realised success rates. Economic return is reported beside this but is not used to redefine the target after the fact.

12. Incremental value and ablation analysis

Increment	Interpretation
Baseline 1 - Baseline 0	Value of econometric relationship validity.
Model A - Baseline 1	Value of supervised quantitative event selection.
Model B - Model A	Value of structured point-in-time event/NLP features.
Full System - Model B	Value of point-in-time language intelligence + agent/controller risk control.

12.1 Feature-family ablations

Ablation	Question
Remove spread-state family	Does the model need immediate divergence dynamics beyond entry z?
Remove relationship-quality family	How much value comes from cointegration/stability diagnostics?
Remove market-context family	Does context improve prediction?
Remove security-state family	Are one-leg momentum/volume shocks informative?
Remove liquidity family	Does implementation state help selection?
Remove event/NLP feature family	How much value comes from structured point-in-time event information?

Ablations are validation-driven

Feature-family decisions are made before final-test opening. The final report can evaluate frozen ablations on test, but should not repeatedly modify the production feature set in response to test results.

13. Agent/controller evaluation

$$\text{Failure Rejection Rate} = \frac{\#\{y = 0 \text{ events withheld}\}}{\#\{y = 0 \text{ Model B events}\}}$$

$$\text{False Veto Rate} = \frac{\#\{y = 1 \text{ events withheld}\}}{\#\{y = 1 \text{ Model B events}\}}$$

Metric	Definition / interpretation
Failed-opportunity rejection rate	Share of Model B failures converted to REVIEW/NO TRADE/ABSTAIN by the full system.
False-veto rate	Share of Model B successes withheld by the full system.
Candidate retention	Share of Model B eligible events that remain CANDIDATE.
Net failure reduction	Change in number/rate of failed simulated trades after agent/controller gate.
Tail-loss reduction	Change in worst/lower-tail event returns and drawdown contribution.
Abstention rate	Share withheld because the system cannot make a defensible classification.
Reason-code distribution	Why opportunities were withheld: high break risk, data failure, low score, market-stress warning, etc.

13.1 Interpretation trade-off

A system that rejects every event has perfect failure rejection but zero usefulness. Agent evaluation must therefore report failure rejection and false-veto/retention together.

14. Human-labelled NLP evaluation subset

The agent layer requires direct evaluation of its evidence-processing quality, not only downstream returns. Build a stratified human-labelled subset from historical events before inspecting full-system test outcomes.

Label field	Human annotation
event_type	Dominant material event category or NONE/UNKNOWN.
affected_leg	A / B / BOTH / UNKNOWN.
direction	Positive / negative / mixed / neutral / unknown.
materiality	LOW / MEDIUM / HIGH / UNKNOWN.
relationship_relevance	LOW / MEDIUM / HIGH / UNKNOWN.
evidence validity	Whether the cited chunk genuinely supports the extracted claim.
source completeness note	Whether available SEC evidence appears sufficient for the label.

- Minimum 50 events; aim for about 100 if group capacity permits.
- Stratify across sectors, years, successful/failed events, and events with/without recent 8-Ks.
- Report per-field accuracy and macro-F1 for categorical fields where class counts support it.
- Report evidence-validity rate separately from event-classification accuracy.

15. Robustness slices and stress assumptions

Slice / sensitivity	Required report
By calendar year	Event count, success rate, mean net return, ML metrics where sample supports.
By sector	Same metrics; flag very small samples.
By pair	Distribution of contribution; identify concentration in a few pairs.
By divergence sign	Positive versus negative z events.
By market stress	Observable VIX / realised-volatility buckets defined before final-test evaluation.
By event type/materiality	Full-system subset where labels exist.
Transaction cost	2.5 / 5 / 10 / 20 bps per side.
Borrow cost	0% / 3% / 10% annualised short notional.
Threshold sensitivity	Validation-focused; report frozen test policy plus nearby pre-specified thresholds as sensitivity, not re-optimisation.
Label sensitivity	Optional pre-specified alternative horizon/exit definitions reported separately; do not choose best after test.

16. Statistical uncertainty and dependence

Divergence events are not perfectly independent: multiple pairs can trigger in the same market shock and the same pair can generate events over time. Standard iid confidence intervals can therefore be overconfident.

Method	Use
Descriptive counts/distributions	Always report N; avoid interpreting tiny slices as stable.
Block bootstrap by calendar month	Resample months, preserving same-month event clustering; construct uncertainty intervals for success rate, mean event return, and selected performance differences.
Pair-cluster sensitivity	Recompute headline event metrics after leaving one pair out or one sector out.
No p-value fishing	Do not run dozens of significance tests and select only favourable ones.

Uncertainty is secondary to design integrity

The most important protection against false discovery is the frozen chronological test and transparent ablation ladder. Bootstrap intervals quantify uncertainty; they do not repair look-ahead or test-set tuning.

17. Error analysis and structural-break case studies

Case type	What to inspect
ML false positive	High P(revert) but y=0: feature state, SHAP, market state, catalyst evidence, adverse path.
ML false negative	Low score but y=1: whether model misses a recurring pattern or feature family.
Correct agent veto	Model B would trade a failure; full system withholds due to validated break evidence.
False agent veto	Model B success withheld; inspect event interpretation and policy threshold.
Agent abstention	Was missing evidence genuinely unavoidable, or is retrieval too weak?
Catastrophic tail event	Trace from raw source -> pair snapshot -> ML -> agents -> controller -> simulated P&L.

17.1 Demo case selection

Choose demo cases after the final evaluation framework is frozen, not by cherry-picking only spectacular wins. Include at least one successful candidate, one correct structural-break rejection, one false veto/review, and one abstention/data-quality case if available.

18. Leakage, multiple testing, and research-governance controls

Risk	Control
Look-ahead prices	Signal at close t, execute at next open; rolling features end at t; labels alone use future.
Future pair parameters	Use snapshot admissible at signal date; never full-sample beta/mean/std.
Future market-state features	Use observable trailing market/VIX/volatility features only.
Future text	SEC acceptance/news publication timestamp <= event cutoff.
Test tuning	Freeze features, model, calibration, thresholds, agent policy and costs before final test.
Pair/data snooping	Curated economic universe; document selection criteria; do not search thousands of pairs until backtest looks good.
Repeated variants	Log every experiment; distinguish pre-specified headline from exploratory sensitivity.
Missing data selection	Report coverage/matched-subset results; do not silently drop difficult full-system events.

Reproducible experiment ID

Every evaluation output should be keyed by experiment_id containing data snapshot, pair/label policy, model A/B versions, agent/policy versions, execution-cost configuration, and code commit.

19. Strategy Lab UI specification

UI panel	Content
Evaluation ladder	Side-by-side table of B0, B1, Model A, Model B, Full System event counts and key outcomes.
Equity curves	Net portfolio curves under identical cost convention; clearly label gross/net and variant.
Predictive diagnostics	ROC/PR, calibration curve, Brier, probability-bucket success rates.
Selection funnel	Raw signals -> valid pairs -> ML accepted -> Model B accepted -> final CANDIDATE, with reasons for rejection.
Agent impact	Failure rejection, false-veto, abstention, break-risk buckets, rejected-trade counterfactual returns.
Robustness controls	Sector/year/market-stress filters; cost/borrow scenario selector.
Trade/event table	Sortable event-level entry/exit, status, P(revert), break risk, return, exit reason and links to Pair Detail.
Experiment manifest	Frozen dates, thresholds, model/policy versions, cost assumptions, data snapshot.

19.1 Visual honesty

- Never default to a cherry-picked sector/year filter.
- Show event counts next to performance metrics.
- Do not hide variants with fewer/missing agent-coverage events; show coverage.
- Separate validation sensitivity charts from final-test headline results.

20. Storage schemas, functions, and artefacts

20.1 DuckDB tables

Object	Grain	Key fields
analytics.raw_divergence_signals	pair x fresh crossing	signal_id, pair_id, date, z, relationship validity, snapshot_id
analytics.backtest_trades	experiment x accepted signal	trade_id, variant, event/signal_id, entry/exit times/prices, weights, costs, returns, exit reason
analytics.backtest_daily	experiment x variant x day	date, variant, NAV, daily return, gross exposure, open count, turnover
analytics.evaluation_metrics	experiment x variant x metric/slice	metric_name, value, N, slice keys, confidence interval if used
analytics.agent_eval	event/subset	human labels, extraction scores, evidence validity, veto outcome
ops.experiments	one experiment	experiment_id, manifest path, code/data/model/policy versions, status

20.2 Core functions

Function	Responsibility
generate_raw_signals(...)	Fresh threshold crossings for Baseline 0 universe.
accept_signal(variant, state)	Apply B0/B1/Model A/Model B/full-system acceptance rule.
simulate_pair_trade(signal, execution_cfg)	Next-open entry/exit, frozen weights, costs, path metrics.
build_portfolio(trades, portfolio_cfg)	Daily NAV/exposure/turnover series.
compute_predictive_metrics(...)	AUC/PR/threshold/calibration statistics.
compute_agent_metrics(...)	Failure rejection, false veto, abstention, evidence metrics.
run_robustness_slices(...)	Sector/year/market-stress/cost/leave-one-out tables.
run_experiment(manifest)	End-to-end frozen evaluation and artefact persistence.

21. Reporting tables and figures

Required output	Minimum content
Table 1 - coverage/funnel	Raw signals, valid signals, ML-complete events, Model A accepted, Model B accepted, full CANDIDATE/REVIEW/NO TRADE/ABSTAIN.
Table 2 - predictive performance	Logistic, Model A, Model B: ROC-AUC, PR-AUC, Brier, precision, recall, F1, N.
Table 3 - event economics	Variant: accepted N, success rate, mean/median gross/net return, MAE, holding days.
Table 4 - portfolio performance	Variant: cumulative/annualised return, annualised volatility, Sharpe, max drawdown, turnover, concurrent positions.
Table 5 - agent impact	Failure rejection, false veto, retention, abstention, evidence validity.
Table 6 - robustness	Key metrics across year/sector/market-stress and cost scenarios.
Figure 1	Evaluation funnel.
Figure 2	Calibration curve / probability-bucket success.
Figure 3	Net equity curves.
Figure 4	Drawdown curves.
Figure 5	Rejected versus accepted event-return distributions.
Figure 6	Agent break-risk / decision confusion-style analysis.

22. Build sequence and acceptance criteria

Order	Deliverable	Acceptance condition
1	Raw signal generator	B0 signal history reproducible independently of pair ACTIVE status.
2	Pair trade simulator	Synthetic fixtures verify direction, beta weights, next-open timing, exits and costs.
3	B0/B1 backtests	Same simulator; only econometric eligibility differs.
4	Model A/B evaluation	Stored frozen predictions drive acceptance; no model fitting inside backtester.
5	Full-system integration	Stored decisions drive trade acceptance; REVIEW/NO TRADE/ABSTAIN withheld.
6	Metrics + portfolio	Event and daily portfolio outputs reconcile to trade records.
7	Agent/human metrics	Labelled subset and veto metrics reproducible.
8	Robustness/bootstrap	Pre-specified sensitivity tables generated from one experiment manifest.
9	Strategy Lab	UI reads stored experiment outputs rather than recomputing ad hoc.
10	Final freeze/run	Final test opened once; report artefacts saved immutably.

- All variants use the same execution and cost engine.
- No final-test threshold/hyperparameter/prompt/policy tuning.
- Every performance number traces to event/trade rows and experiment manifest.
- Coverage and sample counts accompany every metric.
- Headline conclusions distinguish predictive, risk-control, and simulated economic evidence.

Appendix A. Working backtest defaults

Parameter	Default
Signal frequency	Daily close
Entry threshold	fresh |z| >= 2.0
Convergence trigger	|z| <= 0.5
Adverse stop	same-side |z| >= 3.0
Maximum horizon	10 trading days
Execution	Signal after close; enter next open; close-triggered exit next open
Position weights	Frozen beta-weighted pair; gross event notional normalised to 1
Entry cost	5 bps per unit gross
Exit cost	5 bps per unit gross
Borrow	3% annualised short notional baseline
Portfolio leverage	None; total gross <=1.0
Full-system auto-trade status	CANDIDATE only
Model A/B candidate threshold	0.65 working start, validation-frozen
Review band	0.45-0.65 in controller
Final test	2024-2025
Risk-free return in prototype Sharpe	0 unless a separately versioned risk-free series is added; disclose explicitly

Appendix B. Formula reference

$$w_A = -\frac{s}{1+|\beta|}, \quad w_B = \frac{s\beta}{1+|\beta|}$$

$$r_t^{pair} = w_A r_{A,t} + w_B r_{B,t} - c_t$$

$$R_{cum} = \prod_{t=1}^{T}(1+r_t) - 1$$

$$R_{ann} = (1+R_{cum})^{252/T} - 1$$

$$\text{Sharpe} = \frac{\bar{r} - r_f}{\sigma_r}\sqrt{252}$$

$$\text{MDD} = \min_t\left(\frac{W_t}{\max_{s\le t} W_s} - 1\right)$$

$$\text{Turnover}_t = \frac{1}{2}\sum_j |w_{j,t} - w_{j,t-1}|$$

$$\text{Brier} = N^{-1}\sum_i (\hat{p}_i - y_i)^2$$

$$\text{FRR} = \frac{\text{failed events withheld}}{\text{failed Model B events}}, \quad \text{FVR} = \frac{\text{successful events withheld}}{\text{successful Model B events}}$$

References and source basis

Source	Use
Complete Project Overview	Section 13 evaluation ladder, predictive/trading/agent metrics, robustness slices and UI Strategy Lab.
Supervised ML Pipeline specification	Label/event definition, chronological partitions, calibration, threshold policy, SHAP and predictive metrics.
Econometric Relationship Engine specification	Spread/z-score, beta/relationship snapshots, lifecycle validity and event triggers.
Multi-Agent Pipeline specification	Evidence/agent/controller outputs, full-system status semantics and evaluation fields.
Data Architecture & Source Strategy	Adjusted OHLCV and storage/version lineage required for reproducible simulation.
FINS5557 Week 4, Week 6 and Week 8 materials	Canonical trading metrics, transaction-cost backtesting, multi-agent trading evaluation and temporal non-stationarity framing.